EASFL: Energy-Aware, Poisoning-Resilient Federated Learning for Anomaly Detection and Lifetime Extension in Wireless Sensor Networks

Shiva Kumar V and Choupiri Shivakeshi

Abstract

Federated learning (FL) keeps raw sensor data on the nodes of a wireless sensor network (WSN), but exchanging model updates over low-power radios can drain batteries faster than shipping the raw data. We present EASFL, a cluster-based FL framework for on-node anomaly detection that treats energy as a first-class constraint. It combines energy-aware cluster-head and participant selection, top-k sparsified 8-bit updates with error feedback, two-tier robust aggregation with reputation-based quarantine, and zero-communication on-node personalization. In a 100-node simulation with the first-order radio model, EASFL cuts traffic by $13\times$ and energy by $3\times$ relative to hierarchical FedAvg, delays the first node death from round 35 to 547, and at round 200 attains a network-level F1-score of 0.82 against 0.26, keeping F1 within 0.07 of its clean value under four poisoning attacks with 20% malicious nodes. On real readings and measured links from 50 motes of the Intel Berkeley Lab deployment, EASFL matches the accuracy of centralized and federated baselines (F1 0.67 vs. 0.66–0.70) with $20\times$ less traffic than hierarchical FedAvg and keeps every mote alive at round 200. With CC2420 and CC2650 radio parameters it still uses 3.4–$4.8\times$ less energy per round.

Index Terms

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Wireless sensor networks, federated learning, anomaly detection, energy efficiency, network lifetime, clustering, model poisoning, non-IID data, personalization.

I. Introduction

Wireless sensor networks (WSNs) underpin environmental, structural and industrial monitoring in the Internet of Things (IoT). Their readings are routinely corrupted by spikes, stuck-at values, drifts and noise bursts caused by failing hardware, depleted batteries, environmental extremes or tampering [1], [2], and detecting such anomalies close to the source is essential for data quality and security [3]. The conventional approach forwards raw readings to a base station (BS) for centralized learning. This exposes potentially sensitive data, adds latency and, crucially for battery-powered nodes, spends scarce energy on bulk transmission.

Federated learning (FL) [4], [5] offers an alternative: nodes train locally and share only model updates. FL has been applied to IoT anomaly and intrusion detection [6], [7], but most work assumes edge devices with stable power and connectivity [8]. In a WSN, three problems remain. First, *communication energy*: every round moves a full model from every participant, so naive FL can exhaust the network faster than centralized collection (Section VI). Second, *heterogeneity*: each node sees its own environment and fault mix, so data are non-IID and a single global model under-fits individual nodes [9], [10]. Third, *trust*: nodes are physically exposed, and a few compromised nodes can poison the shared model [11], [12]. Existing studies usually address one of these in isolation, and few report how learning choices translate into network lifetime.

This paper proposes EASFL, an energy-aware, poisoning-resilient FL framework that addresses all three jointly within the clustered architecture already standard in WSNs [13]–[15]. Our contributions are:

1) A cluster-based FL protocol whose every component (head selection, participant selection, update compression, aggregation) is chosen to reduce radio energy, with an explicit bit-level cost model.
2) A two-tier robust aggregation scheme with reputation-based quarantine that requires no trusted root dataset, and a zero-communication on-node personalization step that resolves the non-IID accuracy gap.

3) A reproducible evaluation with energy-coupled detection metrics, including a *network-level* F1-score in which nodes that have exhausted their batteries raise no alarms, ablations, four attack models against three robust-aggregation baselines, sensitivity analyses of the synthetic workload, validation with CC2420 and CC2650 radio parameters under IEEE 802.15.4 framing, and an evaluation on real readings and measured links from the Intel Berkeley Lab deployment. All code and seeds are publicly released.

II. Related Work

Energy-efficient clustering. LEACH [13] rotates cluster heads (CHs) randomly to spread load, and HEED [14] uses residual energy and communication cost; see [15] for a survey. These protocols target data gathering; EASFL reuses their structure but adapts head and participant selection to the FL workload.

Communication-efficient and resource-aware FL. Sparsification with error feedback [16], [17] and sparse ternary compression [18] reduce upload volume, client selection accounts for heterogeneous resources [19], hierarchical client–edge–cloud aggregation cuts long-haul traffic [20], and adaptive aggregation frequency trades computation for communication [21]. These works measure cost in bytes or wall-clock time; we measure it in joules under a WSN radio model and report its effect on node deaths.

Robust FL and heterogeneity. Byzantine-robust rules include Krum [11], coordinate-wise median and trimmed mean [22], and FLTrust, which requires a clean root dataset at the server [23]. Adaptive attacks [12] and backdoors [24] show that no single rule suffices. For non-IID data, proximal regularization (FedProx) [25] and personalization [26], [27] are common. Cryptographic secure aggregation [28] protects update confidentiality but not integrity and is orthogonal to our design.

FL for IoT anomaly detection. DÏoT [6] and [7] detect compromised IoT devices with FL, but target gateways or mains-powered devices and do not model radio energy or lifetime. To our knowledge, EASFL is the first framework to evaluate head selection, participant selection, compression, robust aggregation and personalization together under a WSN energy model with lifetime and network-level detection as joint outcomes.

III. System Model

A. Network and Energy Model

N static nodes are deployed uniformly in an $L \times L$ field with a BS outside the field. Node i starts with heterogeneous energy $E_i^0 \sim \mathcal{U}[0.25, 0.75]$ J. Time is divided into rounds; in each round clusters are formed, members exchange data or updates with their CH, and CHs communicate with the BS over a single hop. We adopt the first-order radio model [13]: transmitting ℓ bits over distance d costs

$$E_{\mathrm{tx}}(\ell, d) = \begin{cases} \ell E_{\mathrm{elec}} + \ell \varepsilon_{\mathrm{fs}} d^2, & d < d_0, \\ \ell E_{\mathrm{elec}} + \ell \varepsilon_{\mathrm{mp}} d^4, & d \geq d_0, \end{cases} \tag{1}$$

and $E_{\mathrm{rx}}(\ell) = \ell E_{\mathrm{elec}}$, with $d_0 = \sqrt{\varepsilon_{\mathrm{fs}}/\varepsilon_{\mathrm{mp}}} \approx 87.7\,\mathrm{m}$. CHs spend E_{DA} per bit to aggregate. Local training costs e_s joules per sample-pass, so a round of n_s samples costs $n_s e_s$. Every packet carries a 200-bit header, and cluster advertisement and join messages are 200-bit control packets. A packet is lost independently with probability p_ℓ. A node dies when its energy reaches zero. Lifetime is reported as the rounds until the first node dies (FND), half the nodes die (HND) and 90% die (LND).

B. Sensing Workload and Fault Model

Because labelled fault data from real deployments are scarce and their ground truth is uncertain [2], our main experiments use a fully specified synthetic workload that reproduces the structure of indoor/outdoor environmental WSNs. Node i samples temperature, humidity, light and supply voltage once per minute:

$$\begin{aligned} T_i(t) &= \bar{T}_i + A_i \sin(2\pi t/1440 + \phi_i) + n_T, \\ H_i(t) &= \bar{H}_i - 1.5\,(T_i(t) - \bar{T}_i) + n_H, \\ L_i(t) &= \bar{L}_i \max\big(0, \sin(2\pi t/1440 + \phi_i)\big) + n_L, \\ V_i(t) &= 2.9 - 10^{-5} t + n_V, \end{aligned} \tag{2}$$

with $\bar{T}_i \sim \mathcal{U}[18, 30]^\circ$C, $A_i \sim \mathcal{U}[1, 5]$, $\bar{H}_i \sim \mathcal{U}[30, 60]\%$, $\bar{L}_i \sim \mathcal{U}[100, 600]$ lux and zero-mean Gaussian noise with standard deviations $\kappa_i(0.15, 0.5, 10, 0.005)$, $\kappa_i \sim \mathcal{U}[0.7, 1.6]$; 40% of nodes are outdoor, with three-fold light noise. Readings are grouped into windows of $W = 10$ samples. Each window of node i is anomalous with a node-specific rate

$$\lambda_i = \mathrm{clip}\big(\mathrm{LogNormal}(\ln 0.07, 0.8),\ 0.01,\ 0.30\big),$$

and the fault type is drawn from a node-specific mixture $\boldsymbol{\pi}_i \sim \mathrm{Dir}(\alpha\mathbf{1})$, $\alpha = 0.5$, over the four types catalogued in [1], [2]: *spike* (one sample offset), *stuck-at* (a channel frozen), *drift* (a linear ramp) and *noise burst*. The fault affects one random channel with magnitude $a \sim \mathcal{U}[3, 8]$ times that channel's noise level. The Dirichlet prior makes the label and fault distributions non-IID across nodes, with α controlling the degree [10].

From each window we extract, per channel, the standard deviation, range, maximum absolute first difference, absolute linear-trend slope, maximum absolute residual after detrending and fraction of repeated samples, giving 24 features. Features are scaled by the nominal channel noise, log-compressed and standardized with network-wide means and variances; these 48 scalars are obtained once at deployment by aggregating per-node sums and sums of squares, at a cost comparable to a single model update. Each node holds 200 training windows and is tested on the following 100 windows. Section VI-H repeats the evaluation on real readings.

C. Detector and Threat Model

Every scheme uses the same detector, a 24–16–1 multilayer perceptron with ReLU hidden units and a sigmoid output ($P = 417$ parameters), trained with class-weighted binary cross-entropy (positive weight 4) by mini-batch SGD. In FL rounds, each participant runs one local epoch on up to 150 windows (learning rate 0.15, batch 16) and reports its update $\Delta_i = w_i - w$.

A fraction m of nodes is compromised at deployment. Compromised nodes follow the protocol but corrupt their own updates using one of four attacks: *scaled sign flip* ($-4\Delta_i$); *stealth sign flip* ($-\Delta_i$, which preserves the norm); *label flip* (training on inverted labels); and *Gaussian* (random direction with twice the honest norm). CHs are assumed to aggregate honestly; adversarial CHs are left for future work.

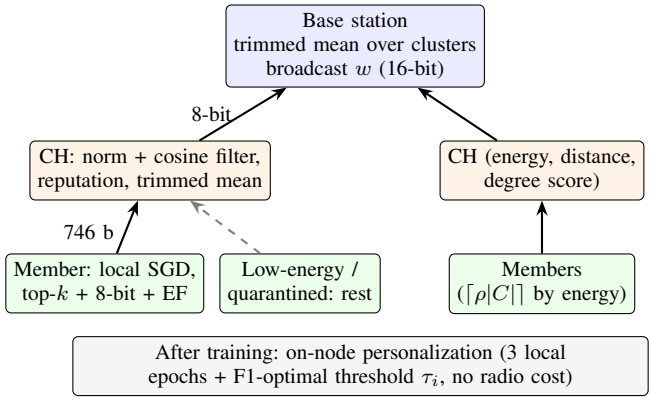


Fig. 1. EASFL round structure. Only selected members train and upload compressed updates; CHs filter and aggregate; the BS aggregates clusters robustly and broadcasts the model.

IV. The EASFL Framework

Fig. 1 and Algorithm 1 summarize one EASFL round.

A. Energy-Aware Cluster-Head Selection

Let $\mathcal{A}$ be the set of alive nodes and $k = \mathrm{round}(p|\mathcal{A}|)$ with $p = 0.1$. Each alive node receives the score

$$S_i = \left(0.5\,\hat{E}_i + 0.3\,\hat{D}_i + 0.2\,\hat{G}_i\right)\mathbf{1}[i \notin \mathcal{Q}], \tag{3}$$

where $\hat{E}_i = E_i / \max_j E_j$ is normalized residual energy, $\hat{D}_i = 1 - (d_i - d_{\min})/(d_{\max} - d_{\min})$ rewards proximity to the BS, $\hat{G}_i$ is the normalized number of neighbours within 25 m, and $\mathcal{Q}$ is the set of quarantined nodes. CHs are chosen greedily in decreasing S_i subject to a minimum separation of 20 m, which spreads clusters spatially. Each member joins its nearest CH. Unlike LEACH's randomized rotation, this keeps the energetically expensive CH role away from weak nodes.

B. Energy-Aware Participant Selection

Within cluster C, only nodes with $E_i > E_{\min} = 0.025\,\mathrm{J}$ that are not quarantined are eligible, and the $\lceil \rho |C| \rceil$ eligible members with the highest residual energy participate ($\rho = 0.5$). The others skip training and uploading but still receive the global model, so they keep detecting with the latest model.

C. Compressed Updates

Participants upload a sparsified, quantized update with error feedback [16], [17]. With residual memory r_i (initially zero), node i forms $v_i = \Delta_i + r_i$, keeps the $K = \lceil 0.1P \rceil = 42$ entries of largest magnitude, quantizes them uniformly to 8 bits with scale $s = \max |v_{i,k}|/127$, sends the result q_i and sets $r_i \leftarrow v_i - q_i$. An upload thus takes $K(8 + \lceil \log_2 P \rceil) + 32 = 746$ bits instead of $32P = 13{,}344$ (17.9× smaller). CHs send dense 8-bit aggregates to the BS (3,368 bits, 4× smaller) and the BS broadcasts the model in 16-bit precision (6,672 bits).

D. Two-Tier Robust Aggregation with Quarantine

At CH c, let $U = \{u_i\}$ be the received (decoded) updates and $\tilde{u} = \mathrm{med}(U)$ their coordinate-wise median. Update u_i is flagged if

$$\|u_i\| > 3\,\mathrm{med}_j \|u_j\| \quad \text{or} \quad \cos(u_i, \tilde{u}) < 0. \tag{4}$$

Each node keeps a suspicion score $f_i \leftarrow 0.7 f_i + \mathbf{1}[\text{flagged}]$; when $f_i > 1.5$ (roughly two flags in three rounds) the node is quarantined for 20 rounds, during which it is excluded from participation and from CH candidacy. Unflagged updates are combined with a coordinate-wise trimmed mean (trim ratio 0.2), or the median if fewer than five remain [22]. The BS combines cluster aggregates with a trimmed mean (ratio 0.1) when at least five arrive, and a sample-weighted mean otherwise. No trusted root dataset is needed, unlike [23].

E. On-Node Personalization

Non-IID fault mixes and rates make a single decision rule sub-optimal for every node. After federated training, each node fine-tunes the received model for three epochs on its own windows (learning rate 0.05) and selects the decision threshold τ_i that maximizes F1 on its training windows, in the spirit of personalized FL [26], [27]. This step costs about 600 sample-passes (≈ 1.2 mJ, the training energy of four FL rounds) and no radio traffic.

V. Experimental Setup

Table I lists the parameters. All results are means ± standard deviations over five seeds; each seed fixes the deployment, energies, attackers and data, so all schemes see identical conditions.

Algorithm 1 One EASFL round

Input: global model w, residual energies $\{E_i\}$, quarantine set $\mathcal{Q}$

Select CHs by score S_i with 20 m separation; members join nearest CH
for each cluster C with head c **do**
  $\mathcal{P}_C \leftarrow$ top-$\lceil \rho|C| \rceil$ by E_i among $\{i \in C : E_i > E_{\min}, i \notin \mathcal{Q}\}$
  for $i \in \mathcal{P}_C$ **do** ▷ in parallel
    $\Delta_i \leftarrow \mathrm{SGD}(w, \mathcal{D}_i) - w$; $v_i \leftarrow \Delta_i + r_i$
    $q_i \leftarrow Q_8(\mathrm{Top}_K(v_i))$; $r_i \leftarrow v_i - q_i$; send q_i to c
  end for
  c: flag by norm/cosine, update f_i and $\mathcal{Q}$, $a_C \leftarrow \mathrm{TrMean}_{0.2}$(unflagged)
  c: send $Q_8(a_C)$ to BS
end for
BS: $w \leftarrow w + \mathrm{TrMean}_{0.1}(\{a_C\})$; broadcast $Q_{16}(w)$

Output: updated w; after training, each node personalizes (w_i, τ_i)

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Nodes, field, BS position	100; 100×100 m; (50, 175) m
Initial energy	$\mathcal{U}[0.25, 0.75]$ J
E_{elec}, E_{DA}	50 nJ/bit, 5 nJ/bit
$\varepsilon_{\mathrm{fs}}$, $\varepsilon_{\mathrm{mp}}$	10 pJ/bit/m^2, 0.0013 pJ/bit/m^4
Training energy e_s	2 μJ per sample-pass
Header / control packet	200 / 200 bits
CH fraction p; participation ρ	0.1; 0.5
Packet loss p_ℓ	0.05 (swept 0–0.3)
Model	MLP 24–16–1, $P = 417$
Local training	1 epoch, $\leq$150 windows, lr 0.15, batch 16
Compression	top-10% (K=42), 8-bit, error feedback
Payloads (bits)	FedAvg 13,344; EASFL 746 up, 3,368 CH$\rightarrow$BS, 6,672 down
Centralized raw data	60 samples $\times$ 4 ch. $\times$ 16 bits per node-round
Rounds; seeds	200 (learning), $\leq$4000 (lifetime); 5

Learning runs last 200 rounds; lifetime runs, which use the same energy accounting, continue until all nodes die (at most 4,000 rounds). The simulator is a round-based Python implementation of the model in Section III; it abstracts the MAC layer and assumes perfect scheduling inside clusters.

Baselines. Centralized (CL): nodes send raw 16-bit samples through LEACH clusters to the BS, which trains the same MLP for 30 epochs on all data. *Local*: each node trains its own MLP for 30 epochs without communication. *FedAvg-flat*: all alive nodes train and send full-precision updates directly to the BS [4]. *FedAvg-LEACH*: hierarchical FedAvg [20] over LEACH clusters with full-precision updates. For robustness we also equip FedAvg-LEACH with coordinate-wise

TABLE II
DETECTION AND EFFICIENCY AFTER 200 ROUNDS (MEAN ± STD OVER 5 SEEDS). G: GLOBAL MODEL; P: PERSONALIZED; NET: NETWORK-LEVEL F1 WITH DEAD NODES SILENT

Scheme	F1 (G)	AUC (G)	FPR (G)	F1 (P)	FPR (P)	Net F1	Traffic (Mbit)	E_{200} (J)	Alive @200	FND
Centralized	0.783±0.021	0.939	0.020	0.812±0.024	0.015	0.699±0.014	137.1	25.9	75.2	25
Local only	0.693±0.031	0.887	0.030	0.708±0.032	0.019	–	0	–	100	–
FedAvg-flat	0.589±0.057	0.908	0.086	0.819±0.017	0.015	0.115±0.041	121.7	49.0	7.6	18
FedAvg-LEACH	0.706±0.042	0.924	0.038	0.823±0.020	0.013	0.258±0.031	197.1	47.3	22.6	35
EASFL	0.711±0.015	0.914	0.030	0.821±0.021	0.013	**0.821**±0.021	**15.2**	**15.6**	**100**	**547**

median, *trimmed mean* (ratio 0.2) [22] and *Krum* ($f = 0.2n$) [11], applied at both CH and BS.

Metrics. Detection: accuracy, precision, recall, F1, false-positive rate (FPR) and AUC over all test windows of all nodes, for the global model (threshold 0.5) and after personalization. Because a depleted node cannot raise alarms, we also report *network-level* F1 at round 200, where every window of a dead node is scored as normal. Efficiency: over-the-air uplink traffic, energy consumed in the first 200 rounds (E_{200}), and FND/HND/LND.

VI. RESULTS

A. Detection, Traffic and Energy

Table II gives the main comparison. Without personalization, the hierarchical FL schemes reach a per-node F1 of about 0.71, below centralized learning (0.78) but above purely local models (0.69). FedAvg-flat is worse (0.59) because its participants die quickly and the model degrades after round 120 (Fig. 2). EASFL matches FedAvg-LEACH in F1 (0.711 vs. 0.706) with a lower FPR (0.030 vs. 0.038) and lower variance across seeds, while transmitting 15.2 Mbit instead of 197.1 Mbit ($13\times$ less) and consuming 15.6 J instead of 47.3 J ($3\times$ less) over 200 rounds.

On-node personalization raises per-node F1 to about 0.82 for every federated scheme, above the centralized global model (0.78) and even the centralized model with per-node thresholds (0.81), and cuts the FPR to 0.013. Personalization is therefore what resolves the non-IID gap; it benefits all FL schemes equally, so EASFL's advantage lies elsewhere. We also tested a FedProx proximal term [25] ($\mu = 0.1$), which gave no gain (F1 0.706) and is not used.

The decisive difference appears at the network level. By round 200 only 7.6 FedAvg-flat nodes and 22.6 FedAvg-LEACH nodes are alive, so most anomalies go unreported and their

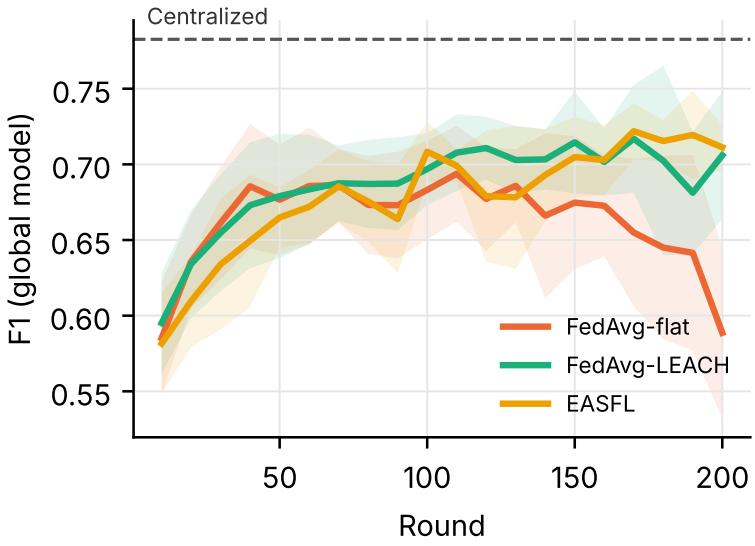


Fig. 2. F1 of the global model during training (mean ± std, 5 seeds). FedAvg-flat degrades as its nodes die.

TABLE III
LIFETIME (ROUNDS) AND ENERGY IN THE FIRST 200 ROUNDS

Scheme / variant	FND	HND	LND	E_{200} (J)
Centralized (LEACH)	25	320	489	25.9
Centralized (EA-CH)	142	246	461	39.1
FedAvg-flat	18	73	189	49.0
FedAvg-LEACH	35	145	233	47.3
EASFL	**547**	645	731	**15.6**
w/o participant selection	317	538	628	19.4
w/o compression	219	248	291	40.9
w/o energy-aware CH (LEACH)	177	**697**	**802**	15.9

network-level F1 falls to 0.12 and 0.26. Centralized collection retains 75 nodes (network F1 0.70). EASFL keeps all 100 nodes alive and its network-level F1 equals its per-node F1, 0.82.

B. Network Lifetime and Ablation

Fig. 3 shows alive nodes over time and Table III the lifetime milestones. EASFL delays the first node death to round 547, against 35 for FedAvg-LEACH (15.6×) and 25 for centralized collection, and its HND (645) is 4.4× that of FedAvg-LEACH. Its characteristic "plateau then cliff" curve reflects balanced drain: no node dies until residual energies have been evened out.

The ablation (Table III, Fig. 4) isolates each component. Compression is the largest contributor:

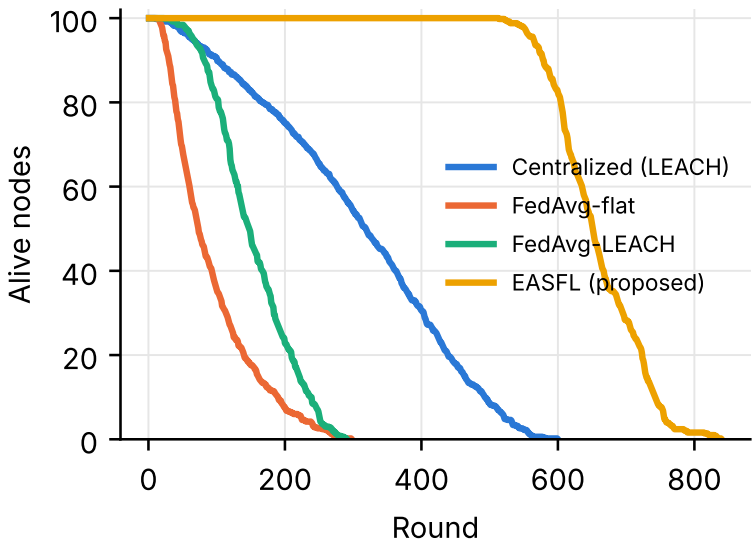


Fig. 3. Alive nodes per round (mean of 5 seeds, 60 raw samples per node-round for centralized collection).

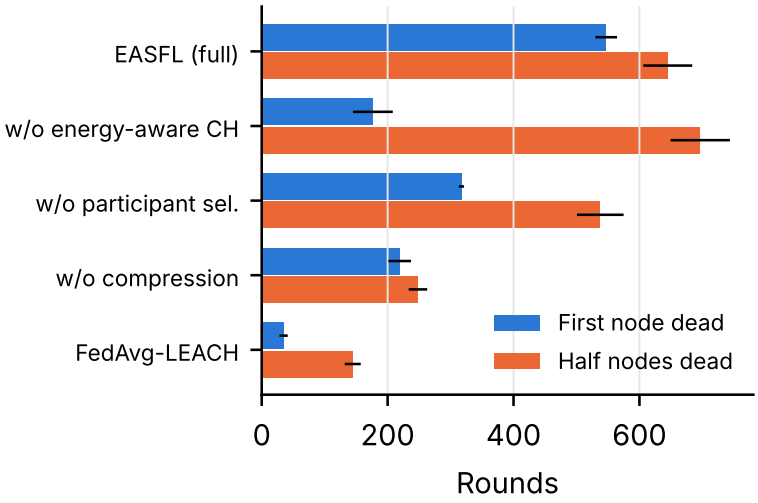


Fig. 4. Ablation: contribution of each EASFL component to lifetime (mean ± std).

without it, energy use rises 2.6× and HND falls from 645 to 248. Participant selection adds 73% to FND. Energy-aware CH selection triples FND (547 vs. 177) but slightly shortens HND and LND relative to LEACH rotation (645 vs. 697, 731 vs. 802). This is a genuine trade-off: deterministic selection favours nodes close to the BS and well connected, which equalizes early drain but concentrates the CH burden on a subset late in life. Applications that need full

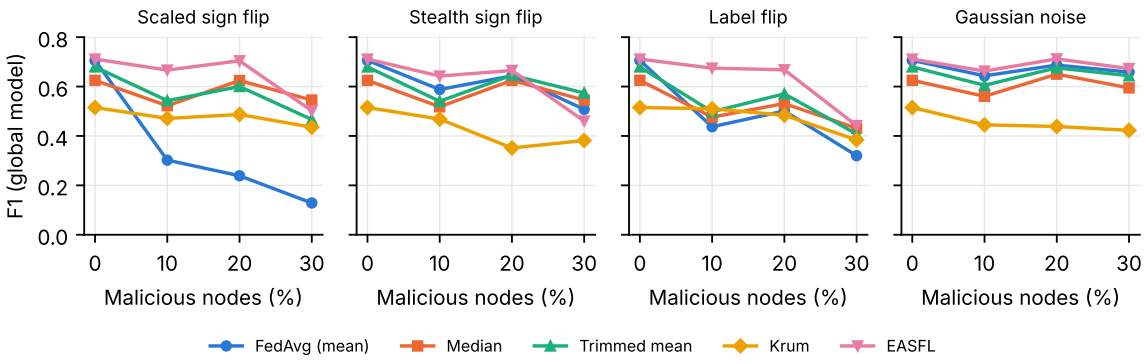


Fig. 5. Global-model F1 versus fraction of malicious nodes for four poisoning attacks (mean of 5 seeds).

coverage (FND) should prefer the energy-aware score; those that tolerate gradual coverage loss may prefer rotation. The same trade-off appears when the energy-aware score is applied to centralized collection (FND 25 $\rightarrow$ 142, HND 320 $\rightarrow$ 246).

C. Robustness to Poisoning

Fig. 5 plots global-model F1 against the fraction of malicious nodes for the four attacks, and Table IV details the 20% case. Undefended FedAvg collapses under the scaled sign flip (F1 0.24 at 20%, 0.13 at 30%). With 10% and 20% attackers, EASFL keeps global F1 between 0.64 and 0.71 across all attacks, within 0.07 of its clean value (0.71) and higher than every compared aggregation rule in all eight cases. Removing EASFL's filtering and quarantine ("EASFL w/o robust") lets the sign-flip attack succeed (0.18), confirming that the defence and not compression or participant selection provides the resilience. The label-flip attack is notable: its updates are neither large nor reversed in direction, yet EASFL (0.67) outperforms median (0.53) and trimmed mean (0.57), because repeated flags quarantine persistent offenders. Krum performs poorly throughout (0.35–0.52), even without attackers, because selecting a single update discards the information carried by heterogeneous nodes.

Two further observations matter for practice. First, personalization is itself a strong mitigation: after local fine-tuning, label-flip and Gaussian attacks barely affect any scheme. It does not, however, rescue the undefended schemes from the sign-flip attack (0.40), so robust aggregation remains necessary. Second, the defence has limits. At 30% attackers EASFL becomes unstable

TABLE IV
F1 WITH 20% MALICIOUS NODES: GLOBAL MODEL / PERSONALIZED

Defence	Sign flip	Stealth	Label flip	Gaussian
FedAvg (mean)	0.24 / 0.40	0.64 / 0.80	0.50 / 0.81	0.69 / 0.83
Median	0.62 / 0.81	0.63 / 0.81	0.53 / 0.82	0.65 / 0.82
Trimmed mean	0.60 / 0.80	0.65 / 0.81	0.57 / 0.82	0.68 / 0.83
Krum	0.49 / 0.75	0.35 / 0.73	0.48 / 0.75	0.44 / 0.73
EASFL w/o robust	0.18 / 0.41	0.66 / 0.81	0.43 / 0.82	0.70 / 0.82
EASFL w/o cosine	0.70 / 0.81	0.66 / 0.80	0.65 / 0.82	0.70 / 0.82
EASFL	**0.70** / 0.82	**0.67** / 0.81	**0.67** / 0.82	**0.71** / 0.82

across seeds (sign flip: F1 0.50 $\pm$ 0.25; stealth: 0.46 $\pm$ 0.21), whereas the coordinate-wise median degrades more gracefully (0.55 in both cases). We therefore claim resilience up to 20% compromised nodes. The cosine test added little beyond the norm test and quarantine (Table IV); it is cheap and retained, but its effect is not statistically distinguishable in our runs.

D. Packet Loss

With packet-loss probabilities from 0 to 0.3, EASFL's global F1 stays between 0.69 and 0.72 and its personalized F1 at 0.82, whereas FedAvg-LEACH's global F1 falls from 0.69 to 0.65 at 30% loss. Because each round aggregates many small, independent uploads, isolated losses mainly reduce the effective participation rather than biasing the update.

E. Sensitivity to the Synthetic Workload

To verify that the conclusions are not artefacts of the generator settings, we varied the Dirichlet concentration $\alpha \in \{0.1, \ldots, 10\}$, the fault magnitude ($0.5$–$1.5\times$) and the anomaly rate ($0.5$–$2\times$). Fig. 6 shows that personalized EASFL matches or exceeds the centralized model in every setting except the smallest faults ($0.5\times$, where all methods reach only F1 ≈ 0.54 because faults are within noise), and stays within 0.02 of personalized FedAvg-LEACH throughout. Energy and traffic do not depend on the data, so EASFL's lifetime advantage is unaffected by these settings.

F. When Is Centralized Collection Preferable?

The energy of centralized collection grows with the raw sampling volume, whereas EASFL's communication does not. Varying the samples each node reports per round, centralized collection

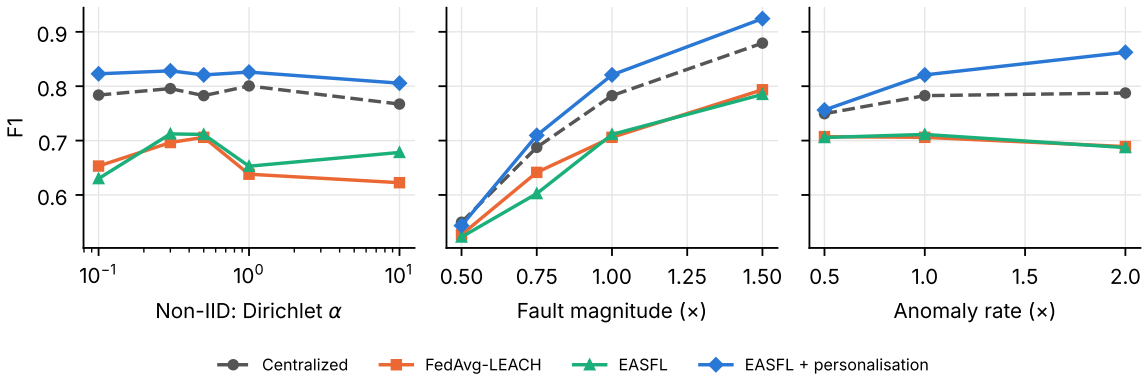


Fig. 6. Sensitivity of F1 to the non-IID degree (Dirichlet α), fault magnitude and anomaly rate (mean of 5 seeds).

with LEACH achieves FND/HND of 81/1379 rounds at 10 samples, 45/589 at 30, 25/320 at 60 and 6/85 at 240. Hence, for very low reporting rates ($\leq$10 samples per round) centralized collection outlives EASFL in HND, although its FND remains far shorter (81 vs. 547); above roughly 30 samples per round EASFL dominates on all lifetime metrics. The model size and the sampling rate together determine which design is preferable, and the cost model of Section III allows this crossover to be computed before deployment.

G. Validation With Commercial Radios

The first-order model charges energy per bit only. To test whether EASFL's gains survive realistic radio behaviour, we replaced it with two IEEE 802.15.4 transceivers at 250 kbit/s and 3.0 V, using datasheet currents: the CC2420 (17.4 mA transmit at 0 dBm, 18.8 mA receive) [29] and the CC2650 (6.1 mA transmit at 0 dBm, 5.9 mA receive) [30]. Every payload is fragmented into frames of at most 102 bytes; each frame adds 17 bytes of PHY and MAC overhead, unicast frames are acknowledged with an 11-byte ACK [31], and each transmission or listening window costs a 1 ms start-up at receive current. Transmit power is fixed, so energy no longer depends on distance. A full-precision update thus occupies 17 frames, the 16-bit downlink 9, and an EASFL member update 2. All other settings are unchanged, and the first-order results are reproduced exactly when the original model is selected.

Table V shows that the ranking is unchanged. With the CC2420, EASFL spends 3.46 mJ per node per round against 12.06 mJ for FedAvg-LEACH ($3.5\times$ less) and delays the first, half and

TABLE V
LIFETIME (ROUNDS) AND MEAN ENERGY PER NODE PER ROUND UNDER THREE RADIO MODELS, 100-NODE NETWORK (MEAN OVER 5 SEEDS). LAST COLUMN: SHARE OF ENERGY SPENT RECEIVING THE MODEL DOWNLINK AND EXCHANGING CLUSTER CONTROL MESSAGES. EASFL-PD: PARTICIPANT-ONLY DOWNLINK

Radio	Scheme	FND	HND	LND	Energy (mJ/node/round)	Downlink + control
First-order	Centralized (LEACH)	25	320	489	2.47	4%
	FedAvg-flat	18	73	189	7.71	9%
	FedAvg-LEACH	35	145	233	3.06	26%
	EASFL	547	645	731	0.79	58%
	EASFL-PD	824	877	937	0.64	48%
CC2420	Centralized (LEACH)	24	139	262	4.22	24%
	FedAvg-flat	34	64	92	7.64	48%
	FedAvg-LEACH	14	42	69	12.06	38%
	EASFL	90	154	174	3.46	84%
	EASFL-PD	181	193	201	2.66	79%
CC2650	Centralized (LEACH)	35	399	783	1.40	23%
	FedAvg-flat	94	179	258	2.72	42%
	FedAvg-LEACH	28	122	200	4.16	35%
	EASFL	285	432	474	1.21	76%
	EASFL-PD	516	541	561	0.96	69%

last node deaths by $6.4\times$, $3.7\times$ and $2.5\times$; with the CC2650 the energy ratio is $3.4\times$ and the first node dies about $10\times$ later. The gains are, however, smaller than under the first-order model (about $16\times$ in FND). Real radios spend at least as much energy receiving as transmitting and pay fixed costs per frame and per wake-up, so the components that every node incurs every round dominate: receiving the broadcast model and the cluster control messages accounts for 84% (CC2420) and 76% (CC2650) of EASFL's energy, whereas member uploads, the target of compression, account for only 4%. Against centralized collection, EASFL still postpones the first node death (90 vs. 24 rounds with the CC2420) but does not win on last node death.

Participant-only downlink (EASFL-PD). This breakdown points to a simple protocol change. Only the nodes that train in a round need the current model, so in EASFL-PD the per-round broadcast is received only by that round's participants and cluster heads, which learn their roles from the cluster set-up messages, while all other nodes sleep through it; every tenth round all nodes receive the model, so that every node's detector stays at most ten rounds old and is current at the evaluation round. Detection is unaffected: over five seeds the global, personalized and network-level F1 change by less than 0.01 on both the synthetic and the real-data workloads. Energy per round falls by 23% with the CC2420 (2.66 vs. 3.46 mJ) and 21% with the CC2650,

and the first node death moves from round 90 to 181 (CC2420) and from 285 to 516 (CC2650), $12.9\times$ and $18.7\times$ later than with FedAvg-LEACH. EASFL-PD still trails centralized collection in last node death (201 vs. 262 rounds with the CC2420, 561 vs. 783 with the CC2650).

Standby current. Real nodes also draw a standby current between rounds. Adding this floor, $VI_{\text{sleep}}T$ per round of period T, the average-node lifetime gain of EASFL over FedAvg-LEACH is $3.1\times$, $2.0\times$ and $1.25\times$ for $T = 1$, 10 and 60 min with the CC2650 (1 μA standby) and $2.2\times$, $1.2\times$ and $1.04\times$ with the CC2420 (20 μA power-down); EASFL-PD raises the 1-min figures to $3.8\times$ and $2.5\times$. The energy savings therefore translate into longer lifetime when learning rounds are frequent or standby currents are low; with hourly rounds on high-leakage hardware the standby floor dominates and the choice of FL protocol matters little for lifetime, although EASFL's advantages in network-level detection and poisoning resilience remain.

H. Real-Data Evaluation: Intel Berkeley Research Lab

To test EASFL beyond the synthetic workload, we used the Intel Berkeley Research Lab trace [32]: 54 Mica2Dot motes in a 40×31 m office sampled temperature, humidity, light and voltage every 31 s, the same four channels as our generator. We used the first ten days (28 February to 9 March 2004), discarded physically impossible readings (temperature outside 10–40 °C, humidity outside 0–100%, voltage outside 2.0–3.2 V) and kept the 50 motes with at least 600 windows of ten consecutive readings. The network uses the real mote positions and the published per-attempt delivery probabilities between motes and to the base station; a unicast is retried up to three times as in 802.15.4, and every attempt costs energy. Members join the cluster head with the best measured link, EASFL's proximity-to-sink term uses the measured link quality instead of distance, and the cluster-head separation and degree radius are scaled to 10 m for the smaller room. Ground-truth labels are obtained by injecting the four fault types into the real readings with the fault process of Section III, with magnitudes in the same absolute units as the synthetic workload. Each seed starts on a different day of the trace; each mote trains on its first 200 windows and is tested on the next 100. The real readings are quantized (consecutive light readings repeat 83% of the time) and contain natural events such as lights switching on, which resemble faults and make the task harder.

TABLE VI
DETECTION AFTER 200 ROUNDS ON THE INTEL BERKELEY LAB DEPLOYMENT (50 MOTES, REAL READINGS, MEASURED LINKS; MEAN ± STD OVER 5 SEEDS). G: GLOBAL MODEL; P: PERSONALIZED; NET: NETWORK-LEVEL F1

Scheme	F1 (G)	AUC (G)	F1 (P)	Net F1	Traffic (Mbit)	Alive @200
Centralized	0.686±0.022	0.917	0.699±0.026	–	–	–
Local only	0.510±0.028	0.815	0.518±0.019	–	0	–
FedAvg-flat	0.697±0.018	0.914	0.691±0.018	0.497±0.044	223.7	29.0
FedAvg-LEACH	0.664±0.017	0.898	0.681±0.026	0.159±0.079	290.3	6.4
EASFL	0.667±0.009	0.904	0.670±0.019	**0.670**±0.019	14.6	50.0

TABLE VII
LIFETIME (ROUNDS) AND ENERGY PER NODE PER ROUND (MJ) ON THE INTEL BERKELEY LAB DEPLOYMENT (MEAN OVER 5 SEEDS)

	First-order			CC2420			CC2650		
Scheme	FND	HND	Energy	FND	HND	Energy	FND	HND	Energy
Centralized (LEACH)	50	775	0.65	18	137	3.88	34	402	1.29
FedAvg-flat	97	220	2.37	21	47	11.05	59	134	3.91
FedAvg-LEACH	54	145	3.23	15	31	16.02	36	84	5.56
EASFL	657	794	0.66	108	159	3.33	342	442	1.17
EASFL-PD	1068	1097	0.51	198	208	2.52	561	578	0.92

Table VI confirms the main findings on real readings. EASFL's global model reaches F1 0.667, on a par with FedAvg-LEACH (0.664) and close to centralized learning (0.686) and FedAvg-flat (0.697), while sending $20\times$ less uplink traffic than FedAvg-LEACH (14.6 vs. 290.3 Mbit). Collaboration matters: models trained only on each mote's own data reach F1 0.51. Because all 50 motes are still alive at round 200, EASFL's network-level F1 equals its per-node F1 (0.670), whereas FedAvg-LEACH, with 6 motes left, detects almost nothing (0.159) and FedAvg-flat reaches 0.497 with 29 motes. Unlike on the synthetic workload, personalization brings no measurable gain here (0.667 vs. 0.670): the motes share one office and their data are far more homogeneous than the strongly non-IID synthetic profiles. EASFL's F1 also has the smallest spread across seeds of all federated schemes.

Table VII shows the same lifetime ranking as in the 100-node network. With the CC2420, EASFL uses $4.8\times$ less energy per round than FedAvg-LEACH (3.33 vs. 16.02 mJ) and postpones the first node death from round 15 to 108; EASFL-PD extends it to round 198. Retransmissions over lossy links penalize schemes that send many frames, so the gap to FedAvg-LEACH is larger than on the synthetic deployment.

With 20% malicious motes, plain FedAvg collapses under the scaled sign-flip attack (F1 0.28), whereas EASFL keeps F1 between 0.58 and 0.65 across the four attacks, the best or within 0.02 of the best robust rule in every case (median, trimmed mean and Krum reach 0.55–0.65).

VII. Discussion and Limitations

Synthetic and injected faults. The synthetic workload provides exact ground truth and controllable heterogeneity, and the Intel Berkeley Lab experiment confirms the main conclusions (comparable accuracy, large traffic, lifetime and network-level gains, resilience to poisoning) on real readings and links. Its labels, however, still come from injected faults, and natural anomalies in the trace are treated as normal data; validation against expert-labelled faults remains open. The personalization gain observed on the synthetic workload did not appear on the more homogeneous real data, so its benefit depends on how heterogeneous the deployment is.

Simulator scope. The round-based simulator abstracts MAC contention, retransmissions and collisions. Section VI-G adds datasheet radio currents, IEEE 802.15.4 framing, acknowledgements, start-up costs and standby current, but contention-based channel access would add further overhead to all schemes, most to those that send the most frames. Training energy uses a fixed per-sample cost; on microcontrollers without floating-point units this cost can be higher, which would favour EASFL's participant selection further.

Threat model. We assume honest CHs and non-colluding, non-adaptive attackers. Adaptive attacks crafted against the specific defence [12], backdoors [24] and compromised CHs remain open. EASFL provides integrity protection, not confidentiality; it can be combined with secure aggregation [28] at the cost of extra communication.

VIII. Conclusion

EASFL shows that federated anomaly detection can extend, rather than shorten, the life of a WSN when every protocol component is designed around radio energy. Compressed, selectively scheduled updates cut traffic by $13\times$ and energy by $3\times$ relative to hierarchical FedAvg and postpone the first node death by more than an order of magnitude; on-node personalization lifts per-node F1 above that of a centralized model without any radio cost; and two-tier robust

aggregation with quarantine maintains accuracy with up to 20% malicious nodes. Measured at the network level, where dead nodes cannot report anomalies, EASFL detects anomalies with F1 0.82 at round 200 while hierarchical FedAvg reaches 0.26. On real readings and measured links from the Intel Berkeley Lab deployment, EASFL matches the accuracy of centralized and federated baselines with $20\times$ less traffic and keeps every mote alive at round 200. With CC2420 and CC2650 radio parameters and IEEE 802.15.4 framing it still uses 3.4–$4.8\times$ less energy per round than hierarchical FedAvg; there the model downlink dominates, and receiving it only at participating nodes (EASFL-PD) cuts energy by a further 21–24% and extends the time to first node death 1.6–$2.0\times$ without affecting accuracy. Standby current reduces the lifetime benefit when rounds are infrequent. Future work will validate EASFL against expert-labelled faults, extend the threat model to adaptive attackers and compromised CHs, and explore adaptive participation ratios driven by energy harvesting.

CODE AVAILABILITY

The simulator, experiment scripts, random seeds and raw results are publicly available at https://github.com/vadlashiva/EASFL.

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